ENGINEERING MATHEMATICS-II UE22MA141B

Properties of Fourier Transform -2

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Fourier Transform of Derivatives

Let $f(t)$ be continuous and $f^{(k)}(t), k = 1,2,\ldots,n$ be piecewise continuous on every interval $[-l, l]$ and $\int_{-\infty}^{\infty} |f^{(k-1)}(t)|\, dt, k = 1,2,\ldots,n$ converges. Let $f^{(k)}(t) \to 0$ as $t \to \infty$ for $k = 0,1,2,\ldots,n-1$. If $\mathcal{F}[f(t)] = F(\omega)$, then

$$\mathcal{F}\left[f^{(n)}(t)\right] = (i\omega)^n F(\omega),$$

Where $f(t)$ and all its derivatives vanish at infinity.

For Example,

$$\mathcal{F}[f'(t)] = \int_{-\infty}^{\infty} f'(t)e^{-i\omega t}dt$$

$$= \left[f(t)e^{-i\omega t}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} i\omega f(t)e^{-i\omega t}dt = i\omega F(\omega).$$

$$\mathcal{F}[f''(t)] = -\omega^2 F(\omega).$$

Example

Solve differential equation using Fourier transform

$$f' - 4f = u(t)e^{-4t}$$

$$i\omega F(\omega) - 4\mathrm{F}(\omega) = \frac{1}{4 + i\omega}$$

$$(i\omega - 4)F(\omega) = \frac{1}{4 + i\omega}$$

$$F(\omega) = \frac{1}{(4 + i\omega)(i\omega - 4)}$$

$$F(\omega) = \frac{-1}{(4 + i\omega)(4 - i\omega)}$$

$$F(\omega) = -\frac{1}{4^2 + \omega^2}$$

$$\left\{\frac{2\text{a}}{a^2 + \omega^2} = \mathcal{F}\left[e^{-a|t|}\right]\right\}$$

$$f(t) = -\frac{1}{2*4}e^{-4|t|} = -\frac{1}{8}e^{-4|t|}$$

Symmetry property of Fourier transforms

Let $\mathcal{F}[f(t)] = F(\omega)$. Then

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$f(t) = \mathcal{F}^{-1}[\omega] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{i\omega t}\,d\omega$$

$$2\pi\, f(t) = \int_{-\infty}^{\infty} F(s)e^{ist}\,ds$$

$$2\pi\, f(-\omega\,) = \int_{-\infty}^{\infty} F(s)e^{-is\omega}\,ds \quad change\;\; s \to t$$

$$= \int_{-\infty}^{\infty} F(t)e^{-i\omega t}\,dt = \mathcal{F}[F(t)]$$

Example

Find Fourier transform of $\frac{1}{2\pi} sinct$

Solution

We know that $F(\omega) = \frac{1}{\sqrt{2\pi}} sinc\omega \quad for$

$$f(t) = \begin{cases} 0 & t < -1 \\ 1 & -1 < t < 1 \\ 0 & t > 1 \end{cases}$$ typical box wave

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$\mathcal{F}\left(\frac{1}{\sqrt{2\pi}} sinct\right) = \mathcal{F}\big(F(t)\big) =$$

$$2\pi f(-\omega)=\begin{cases} 0 & -\omega < -1 \\ 2\pi & -1 < -\omega < 1 \\ 0 & -\omega > 1 \end{cases}$$

Differential with respect to frequency (ω)

If $\mathcal{F}\big(f(t)\big) = F\ (\omega)$ then

$$\mathcal{F}\big(t^n f(t)\big) = i^n F^{(n)}(\omega)$$

Differentiating with respect to ω

$$F'(\omega) = \frac{d}{d\omega}\left(\int_{-\infty}^{\infty} f(t)\,.\,e^{-i\omega t}\,dt\right)$$

$$F'(\omega) = \int_{-\infty}^{\infty} \frac{d}{d\omega}\left(f(t)\,.\,e^{-i\omega t}\right)dt$$

$$F'(\omega) = \int_{-\infty}^{\infty} -it.f(t).e^{-i\omega t}\,dt$$

$$F'(\omega) = -i[\mathcal{F}(tf(t))]$$

Similarly

$$F''(\omega) = (-i)^2[\mathcal{F}(t^2 f(t))]$$

Therefore $\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$

$$(-i)^n t^n f(t) = \mathcal{F}^{-1}(F^{(n)}(\omega))$$

Example

Obtain Fourier transform for $[te^{-4t}u(t)]$.

solution : w.K.t $\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$

And from previous property $\mathcal{F}\left(t^n f(t)\right) = i^n F^{(n)}(\omega)$

$$\mathcal{F}[e^{-4t}u(t)] = \frac{1}{4+i\omega}$$

$$\mathcal{F}[te^{-4t}u(t)] = i.\frac{d}{d\omega}\left[\frac{1}{4+i\omega}\right] = i\left[-\frac{i}{(4+i\omega)^2}\right] = \frac{1}{(4+i\omega)^2}$$

Fourier transform of basic functions:

Unit impulse function, $f(t) = \delta(t)$

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t)\,.\,e^{-i\omega t}\,dt = \left[e^{-i\omega t}\right]_{t=0} = 1$$

∴ Impulse in time ⇒constant in frequency

Fourier transform of Constant function,
$f(t) = k$

By Duality property, we have

If $\mathcal{F}[f(t)] = F(\omega)$ then $\mathcal{F}[F(t)] = 2\pi f(-\omega)$

We know that $\mathcal{F}[\delta(t)] = 1$

$\Rightarrow \mathcal{F}[1] = 2\pi\delta(-\omega) = 2\pi\delta(\omega)$ ($\because$ impulse function is even)

$$\therefore \mathcal{F}[k] = 2\pi k\delta(\omega)$$

Hence constant in time $\Rightarrow$ Impulse in frequency

1. $f(t) = e^{iat}$

By frequency shifting property, we have

$$\mathcal{F}\left[e^{iat} f(t)\right] = F(w-a)$$

Also $\mathcal{F}[1] = 2\pi\delta(\omega)$

$$\therefore \mathcal{F}\left[1. e^{iat}\right] = \mathcal{F}\left[e^{iat}\right] = 2\pi\delta(w-a)$$

1. Trigonometric functions, $f(t) = \cos at$ and $f(t) = \sin at$

$$\mathcal{F}[\cos at] = \mathcal{F}\left[\frac{e^{iat} + e^{-iat}}{2}\right] = \frac{1}{2}\{\mathcal{F}[e^{iat}] + \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\cos at] = \frac{1}{2}\{2\pi\delta(\omega - a) + 2\pi\delta(\omega + a)\}$$

$$\therefore \mathcal{F}[\cos at] = \pi\delta(\omega - a) + \pi\delta(\omega + a)$$

$$= \pi\{\delta(\omega - a) + \delta(\omega + a)\}$$

$$\mathcal{F}[\sin at] = \mathcal{F}\left[\frac{e^{iat} - e^{-iat}}{2i}\right] = \frac{1}{2i}\{\mathcal{F}[e^{iat}] - \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\sin at] = \frac{1}{2i}\{2\pi\delta(\omega - a) - 2\pi\delta(\omega + a)\}$$

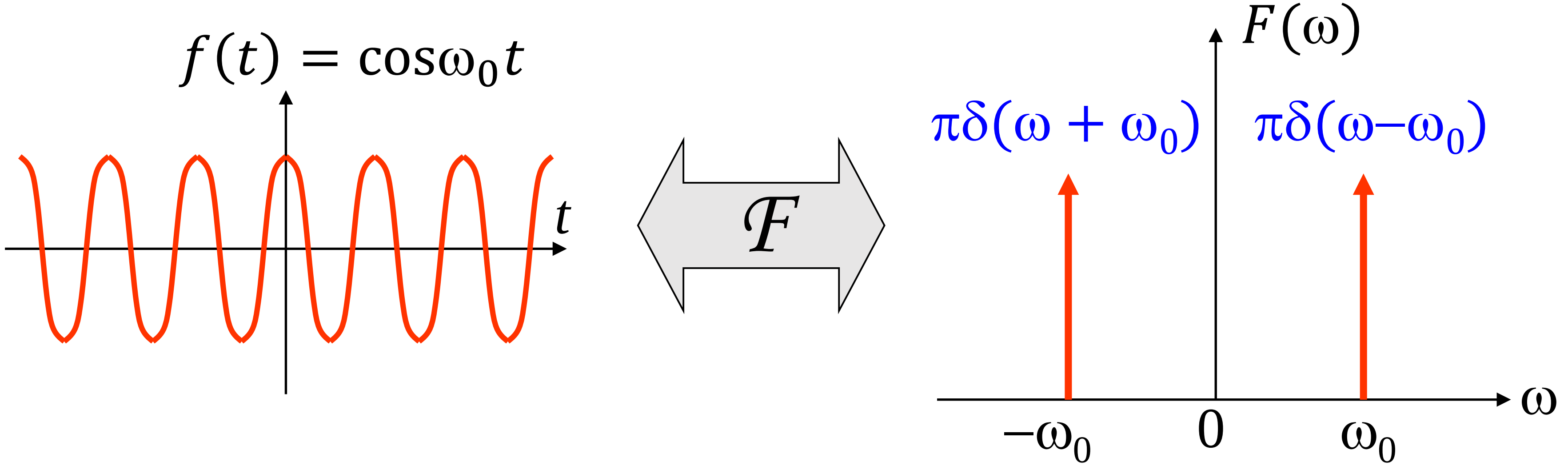
$$= \frac{1}{i}\{\pi\delta(\omega - a) - \pi\delta(\omega + a)\}$$

$$\therefore \mathcal{F}[\sin at] = -i\pi\delta(\omega - a) + i\pi\delta(\omega + a) = i\pi\{\delta(\omega + a) - \delta(\omega - a)\}$$

Fourier Transforms of Sinusoidal Functions

$$\cos \omega_0 t \xleftrightarrow{\mathcal{F}} \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

$$\sin(\omega_0 t) \xleftrightarrow{\mathcal{F}} -i\pi\delta(\omega - \omega_0) + i\pi\delta(\omega + \omega_0)$$



- Linearity property:

$$\mathcal{F}[a\, f(t) + b\, g(t)] = a\, \mathcal{F}[f(t)] + b\, \mathcal{F}[g(t)].$$

- Shifting property: $\mathcal{F}[f(t-a)] = e^{-i\omega a} F(\omega)$
- Frequency shifting property: $\mathcal{F}\left[e^{iat} f(t)\right] = F(w-a)$
- Modulation Theorem: $\mathcal{F}[f(t)\cos(\omega_0 t)] = \frac{1}{2}[F(\omega+\omega_0) + F(\omega-\omega_0)]$
- Fourier Transform of Derivatives: $\mathcal{F}\left[f^{(n)}(t)\right] = (i\omega)^n F(\omega),$
- Differential with respect to frequency (ω):$\mathcal{F}\left(t^n f(t)\right) = i^n F^{(n)}(\omega)$
- Symmetry property $\mathcal{F}[F(t)] = 2\pi f(-\omega)$

- $\mathcal{F}[\delta(t)] = 1$
- $\mathcal{F}\left[e^{iat}\right] = 2\pi\delta(w - a)$
- $\mathcal{F}[\cos at] = \pi\{\delta(\omega - a) + \delta(\omega + a)\}$
- $\mathcal{F}[\sin at] = -i\pi\{\delta(\omega + a) - \delta(\omega - a)\}$
- $\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$
- $\mathcal{F}[e^{-a|t|}] = \frac{2a}{a^2+\omega^2}$

Thank you

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